3.4.5	**Compounds Containing the Carbonyl Group**	
	Aldehydes and ketones	know that aldehydes are readily oxidised to carboxylic acids and that this forms the basis of a simple chemical test to distinguish between aldehydes and ketones (e.g. Fehling's solution and Tollens' reagent)
		appreciate the hazards of synthesis using HCN/KCN
		know that aldehydes can be reduced to primary alcohols and ketones to secondary alcohols using reducing agents such as $NaBH_4$. Mechanisms showing H^- are required (equations showing [H] as reductant are acceptable)
		understand the mechanism of the reaction of carbonyl compounds with HCN as a further example of nucleophilic addition producing hydroxynitriles
	Carboxylic acids and esters	know that carboxylic acids are weak acids but will liberate CO_2 from carbonates
		know that carboxylic acids and alcohols react, in the presence of a strong acid catalyst, to give esters
		know that esters can have pleasant smells
		know the common uses of esters (e.g. in solvents, plasticizers, perfumes and food flavourings)
		know that vegetable oils and animal fats are esters of propane-1,2,3-triol (glycerol)
		know that esters can be hydrolysed
		understand that vegetable oils and animal fats can be hydrolysed to give soap, glycerol and long chain carboxylic (fatty) acids
		know that biodiesel is a mixture of methyl esters of long chain carboxylic acids
	Acylation	know that vegetable oils can be converted into biodiesel by reaction with methanol in the presence of a catalyst
		know the reaction with ammonia and primary amines with acyl chlorides and acid anhydrides
		know the reactions of water, alcohols, ammonia and primary amines with acyl chlorides and acid anhydrides
		understand the mechanism of nucleophilic addition–elimination reactions between water, alcohols, ammonia and primary amines with acyl chlorides
		understand the industrial advantages of ethanoic anhydride over ethanoyl chloride in the manufacture of the drug aspirin